Rutgers University
Fall 2021
Math 151 Sections 54-56

Workshop 7

Problem 1
Keywords: Implicit Differentiation, Tangent Lines

Consider the curve of equation $e^{x^2 y} = 2x + 2y$.

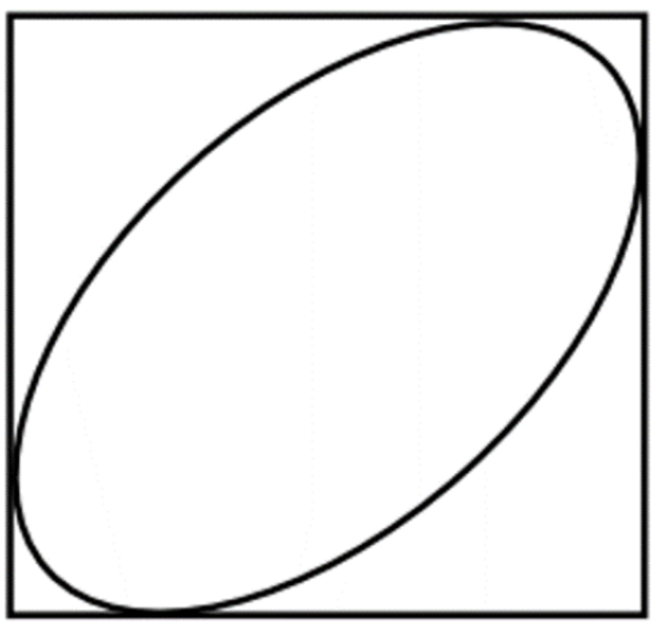
(a) Compute $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ on the curve.

(b) Find the point(s) of intersection of the curve and the x-axis, and the point(s) of intersection of the curve and the y-axis.

(c) Find an equation of the tangent line to the curve at the points you found in the previous question.

Problem 2
Keywords: Implicit Differentiation, Horizontal and Vertical Tangent Lines

The graph of the tilted ellipse $x^2 - xy + y^2 = 3$ is shown below. The sides of the box containing are vertical and horizontal, and are also tangent to the ellipse.



(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ for the tilted ellipse.

(b) Find the points on the ellipse where the tangent line is horizontal.

(c) Find the points on the ellipse where the tangent line is vertical.

(d) What are the dimensions of the box containing the ellipse?

(e) What are the coordinates of the four corners of the box containing the ellipse?

Problem 3

Keywords: Implicit Differentiation, Tangent Lines with a Given Slope

Consider the ellipse of equation $x^2 - 2xy + 3y^2 = 44$.

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ for this ellipse.

(b) Find all the points on the ellipse where the tangent line has slope 3.

(c) For what values of the constant c is the line of equation $y = 3x + c$ tangent to the ellipse? For each value of c, say at which point of the ellipse the line is tangent.